Rubber Molded End Filters

- **Rugged Rubber End Seals**
  - No Bypass. No Cracking w/Age.
- **Textile Media -** Not Paper
  - Handles Moisture, Vibration, Abuse.
- **Genuinely Cleanable**
  - Practical & Economical
- **Exceptional Performance**
  - Lower ΔP, Longer Life

More than 50 years ago, molded end filter elements challenged the worst of punishments in industrial and military services. Today, with synthetic rubber ends, they are arguably the finest air/gas filters ever made.

We manufacture a broad range of sizes …Overall heights to 40", outside diameters to 36", and inside diameters from 1" to 30". *(If you need larger, please call for specific information.)* They can handle air/gas flows to 20,000 CFM. Some provide particle retentions down to 99.9% at 0.1 µ (micron).

These molded end filters are cylindrically shaped. They are designed to cover an intake opening, being held in place by a reuseable plate fitting over a center rod assembly. They have solid rubber ends, heavy duty perforated mild steel, 304SS or 316SS center cores, and radially pleated filter media jacketed with woven wire screen. This screen jacket holds fins open, greatly improving flow and life. Their textile media are well known for superior performance and cleanability vs. fragile paper media. This rugged construction has been long proven to yield higher flows, longer life, and lower ΔP!

New urethane rubber molding systems yield superior filter end seals at production run economy. Don't settle for PVC ends that can soften at intermittent elevated temps., or crumble if undercured.

Rubber Molded End Style - *Cylindrical, double open ends (DOE) of polyurethane rubber, perforated or expanded metal center core, pleated textile, or wire screen filter media. ODs 3" -36", OVHTs 2" - 40", Pleat depths from 0.5" - 4.25". These are arguably the finest built, self sealing, low micron, low ΔP, high flow, field cleanable, and fool proof filter elements available today.*

Rubber Molded Ends.

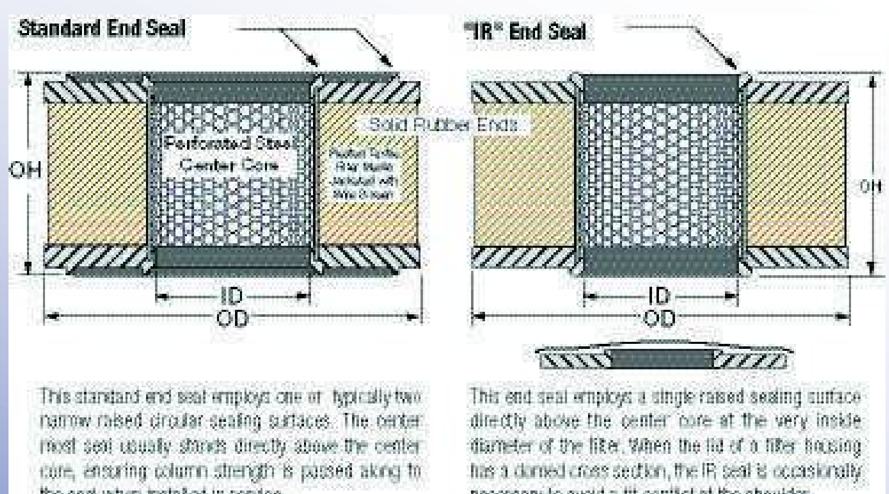
The filter media, support screens, and element cores are bonded together by synthetic rubber ends. Rubber is much more rugged and durable- than lesser polyvinyl elastomers used by others. Rubber will not crumble in service as undercured PVC can. PVC is as much as 50% plastisizer. When this plastisizer evaporates, PVC ends can crack and fail. Our standard synthetic rubber ends are black. We offer colored rubber as well, such as white for food service applications.

We carve production molds from solid stock (no stampings) for exacting end seals. These seals stop dirt cold, and resist most oils and solvents, moisture, or vibrational punishment. They can withstand contiuous service to 250° F, and intermitten service at 350°F*. Optional Silicone rubber ends can serve to nearly 500˚F.

**Performance considerations vary with elevated service temperature and environments. Metal end options with high temp potting materials can serve to over 2000°F.*

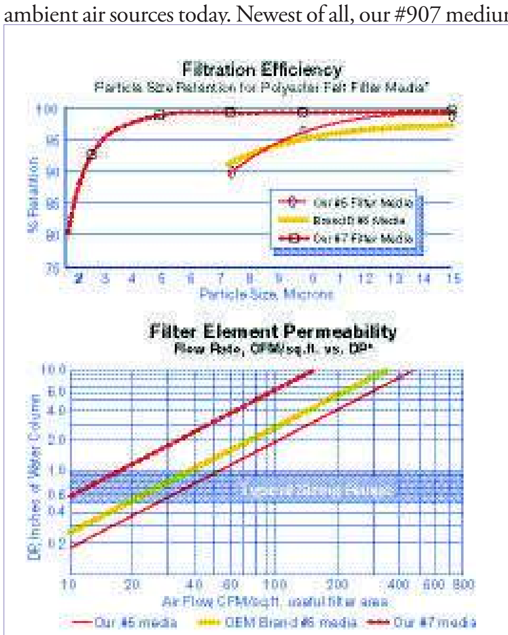
An Element's Core Is Its Heart.

A filter with a weak center core is a house built of straw. We routinely use 16 and 20 gauge perforated steel with 58% open area for low ΔP and high column strength to support the innermost end seal. We weld metal core seams and use premium corrosion inhibitors that will not flake off as paint can. These elements are designed to stand up to abuse.



From left to right above, SEWN END, ACCORDION, RUBBER MOLDED END, and PANEL filter elements. Each style can be supplied with different filter media, and other variations. Our filter elements surpass the most stringent requirements for long life, low ΔP, positive seals, and maximum air flow in compact and cleanable units. Our molded end filter elements do not require a bothersome expanded metal outer wrap to prevent the handling damage common to lesser paper filter media. Instead, we pleat textile filter media between layers of epoxy coated wire screen to yield a rugged jacketed media with 1/4th the resistance to flow of paper media. Jacketed fins resist collapse, have exceptionally high flow, long life (a year is common), and are unharmed by moisture, vibration, pulse flow, and most other service hazards. This also simplifies cleaning with air guns or spray cleaning units. Element cores are 58% open perforated steel. These cores retain column strength where lesser expanded metal, or woven wire cores fail. Our molded urethane rubber ends out perform lesser molded PVC ends offered by many competitors.

Filter media: (see table) #5 polyester felt is arguably the most rugged, washable, 10μ media ever offered. Our #7 polyester medium stops 4μ particles. Elements with 2μ #51 fiberglass are rugged, but not washable. Our 0.1μ HEPA grade #904 medium can stop bacteria. Our #916 medium has 50% activated carbon and can strip away undesirable vapors. Our #910 medium outshines other low cost alternatives at stopping the airborne lint and dirt prevalent in ambient air sources today. Newest of all, our #907 medium with reverse flow radial fin design effectively coalesces smoke and mists without high ΔP loss!



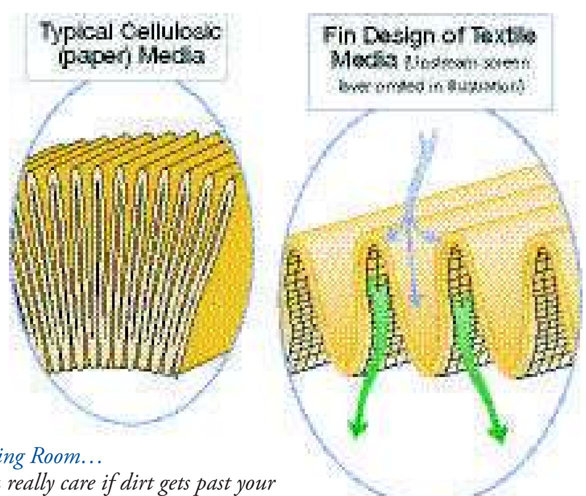
* Instrumentation- HIAC 4100/1100 sensor;
Efficiency counts/250 cu.ft.

Optional Filter Media

Media Suffix #	Filter Media Description:	Reten.* μ Liq	Reten.* μ Gas	Temp. °F	Style* used in:
1	Woven Cotton	30	≤ 30	200	S , M
2	Rayon Felt	> 10	10	200	S , M
3	Woven Cotton	5	2	200	S , M
5	Polyester Felt	20	10	300	S , M
7	Polyester Felt	5	4	300	S , M
8	Woven Fiberglass	15	10	700	S , M
12	Cotton Terrycloth	20	≤ 20	200	S , M
26	304 SS, 100 mesh	150	150	1000	M
30	304 SS, 200 mesh	75	75	1000	M
42	Woven Cotton	1	Not Rated	200	S , M
47	304 SS, 325 mesh	40	40	1000	M
51	Fiberglass Felt, Yellow	> 1	1	450	M
59	Woven Nylon	5	≤ 5	250	S , M
60	Woven Nylon	45	≤ 45	250	S , M
61	304SS, 200 x 1400 mesh	15	15	1000	M
62	304SS, 325 x 2300 mesh	10	10	1000	M
63	Fiberglass Locked/Felt	> 1.5	1.5	500	S , M
64	Polyester Felt	5	4	300	S , M
65	Woven Nylon	90	≤ 90	250	S , M
66	Woven Polyester	2	Not Rated	300	S , M
69	Dynel, woven	2	Not Rated	200	S , M
72	Polyester Felt	2	2	300	S , M
85	Woven Teflon®	10	≤ 10	450	S , M
86	Teflon Felt	10	5	450	S , M
90	Polyester Felt	Not Rated	40	300	M
99	Polyester Felt - Now a misnomer. Depending upon OEM brand, is either #5 or #7 media. Order #7 media if 4μ is needed.				
100	Woven Polypropylene	15	10	175	S , M
101	Woven Polypropylene	10	5	175	S , M
102	Woven Polypropylene	5	3	175	S , M
103	Woven Polypropylene	1	1	175	S , M
105	Fiberglass Felt, Pink	> 2	2	450	M
108	Fiberglass Felt, Pink	> 0.3	0.3	450	M
111	304SS, 50 mesh	280	280	1000	M
135	Woven Fiberglass	6	3	700	S , M
139	Nomex Felt	10	5	450	S , M
142	Polypropylene Felt	10	5	175	S , M
169	Polyester Felt	20	10	300	S , M
200	Galv. C.S. mesh	750	750	500	M
212	Rayon/Nylon Felt	50	50	200	S , M
214	Rayon/Nylon Felt	100	100	200	S , M
418	Woven Polyester	75	75	300	S , M
703	Woven Virgin Teflon®	10	10	450	S , M
704	Woven Polyester	10	8	300	S , M
900	Paper / Microglass	0.5	0.3 abs	180	M
904	Microglass	0.5	0.1	400	M
906	Microglass combination	> 1	1	200	M
907	Microglass combination	> 0.3	≤ 0.3	200	M
910	Polyester/Cotton Felt	> 40	40	300	M
916	Activ. Carbon/Glass	Not Rated	Not Rated	200	M
920	Treated Microglass	Not Rated	1.5	400	M
921	Poly/Glass	Not Rated	0.1	200	M
923	Polypropylene	Not Rated	25	175	M
924	Poly/Glass	Not Rated	< 0.3	200	M
926	Poly/Glass	Not Rated	< 0.3	200	M
927	Poly/Glass	Not Rated	< 0.3	200	M
928	304SS mesh, 50 x 200	Not Rated	60	700	M
931	PTFE Finished Microglass	Not Rated	4	500	S, M
932	Polyester Felt	40	25	300	S, M

Call Us For Many Additional Special Purpose Filter Medium

* S = Sewn End, M = Molded End



Breathing Room…

Do you really care if dirt gets past your filter? Is it worth trying to save a buck on paper rather than rugged textile media? Oddly enough, paper elements cost much more in the long run. Paper pleats crack where you can't see. The light bulb trick won't reveal the failure(s) either. Moisture can ruin paper. And, be very careful of vibration or handling damage. Elements with high performance textile media benefit from 1/3rd the resistance to flow of paper media. They allow open pleat spacing, higher dirt holding capacity, are practical to clean, have lower ΔP, and longer life. Rugged polyester felt media won't crack, tolerates being soaking wet, and takes a beating. Protect your equipment, use textile media.

Coalescing Filter Elements

Sparksfilters coalescing elements can eliminate mists and particulates from your air/gas pipelines. We offer several media options that are up to 98% efficient down to 0.1 Microns. We also offer media and core options that are compatible in environments with more aggressive chemistry.

Before and after, shows one of our coalescing elements and the pipe scale and dirt it filtered out of an air line. (Coalescing flow is inside to outside) At right is a new element ready to go into service

Smoke/Mist Eliminators

These filter elements directly replace the non-pleated elements offered by others. They are intended to remove visible mist & smoke from vacuum pump discharges and other pipeline services. They also serve as fine breather elements at compressor, turbine or other tank reservoir vents. We offer not only this *concentric cylinder* design, but also the higher performance newer low ΔP extended area u …see our series H filter housings on page 28. The modern eliminators (elements shown immediately above) offer sm mist elimination in a smaller and more efficient size.

Wire Mesh Filters

We Call These Bird Screens … When filtration of visible airborne lint and other large particles is all that is required, and little or no ΔP is tolerable, Wire Mesh Filters are commonly used to knock down the boulders. Reportedly, 90+% of particles 10μ and larger are viscously impinged onto the layers of oil wetted crimped wire mesh. (you provide the oil) These elements can be field cleaned and re-oiled. They provide a very minimum degree of filtration.

We replace metal ended OEM filters or those with no ends at all with our rugged self gasketing urethane rubber ends at no extra cost. No gaskets to fool with & a nice fit. We can also supply end plates or end caps to meet OEM specifications.

Experience has shown that the air intake filter is perhaps the most important item ensuring the problem free operation of engines, compressors and blowers! Keep in mind that woven wire mesh media is not a very effective filter media. If you'd rather have a real filter with 10μ polyester felt … please call.

Absorption Pad Sets

Standard absorption media, felt pads, absorb up to 75% of their weight of oil or w also serve as depth filters with retention to 1μ. Each catalog number is supplied a The final number(s) in many competitor's cross references usually indicate the numb per set.

Panel Filters

For Compressor, Blower, and Turbine Intakes

- **Interchange w/Orig. Equip.**
- **Rugged Construction**
- **Significant $avings**
- **Handling Straps**
- **Face Guards**
- **Long Life**

These panels have proven their exceptional performance and value in replacing existing panel filters. Unlike HVAC filters, these high efficiency filters efficiently remove the very fine dust and dirt particles your eyes can't see, from 25 to 0.3 micron… protection required to serve engines, blowers, axial compressors, and turbines… with high efficiency, and low ΔP.

OEM Cat#	OEM Brand	Sparks™ Cat#	OEM Cat#	OEM Brand	Sparks™ Cat#	OEM Cat#	OEM Brand	Sparks™ Cat#
11293	IFM	329-9928	45005	Snshn	329-9927	CP4364	IFM	329-9927
12005	IFM	329-9929	45006	Snshn	329-9925	P0540009-189	Joy	329-9928
12099	IFM	329-9929	45007	Snshn	329-9926	P0540009-190	Joy	329-9927
12164	IFM	329-9927	45008	Snshn	329-9927	P0540009-361	Joy	329-9926
12166	IFM	329-9925	45010	Snshn	329-9925	P0540009-362	Joy	329-9925
15752	Cnslr/Graver	329-9925	45022	Snshn	329-9876K5	P0540009-491	Joy	329-9929
15753	Cnslr/Graver	329-9926	78176	IFM	329-9926	P151	IFM	329-9876
15754	Cnslr/Graver	329-9927	1X2876	IR	329-9928	P20252-90	IFM	329-9876K90
15755	Cnslr/Graver	329-9928	1X2877	IR	329-9927	PF2102100	Nafco	329-9929
15800	Cnslr/Graver	329-9929	1X5573	IR	329-9928	PF2201100	Nafco	329-9928
15860	Cnslr/Graver	329-9929	1X5575	IR	329-9926	PF2212020	Nafco	329-9927
15937	Cnslr/Graver	329-9928	1X5577	IR	329-9929	PF2302040	Nafco	329-9929
16100	Cnslr/Graver	329-9925	1X6410	IR	329-9929	PF2306020	Nafco	329-9926
16101	Cnslr/Graver	329-9926	1X7169	IR	329-9925	PF2312003	Nafco	329-9925
16102	Cnslr/Graver	329-9927	1X8258	IR	329-9928	VE-1103-2424-093	Dlngr	329-9928
18008	Cnslr/Graver	329-9929	1X8259	IR	329-9875	VE-1113-2424-005	Dlngr	329-9929
20523	Cnslr/Graver	329-9929	1X8260	IR	329-9929	VE-1113-2424-099	Dlngr	329-9929
45001	Snshn	329-9928	1X9589	IR	329-9929	VE-1304-2424-176	Dlngr	329-9926
45002	Snshn	329-9928	81-0165	Univ	329-9876K200	VE-1305-2424-164	Dlngr	329-9927
45003	Snshn	329-9929	81-0173	Univ	329-9876K5	VE-1305-2424-166	Dlngr	329-9925
45004	Snshn	329-9929	CP4093	IFM	329-9928	VE-1315-2424-178	Dlngr	329-9927
45005	Snshn	329-9875	CP4166	IFM	329-9925	VE-3305-2424-164	Dlngr	329-9927
45009	Snshn	329-9929	CP4276	IFM	329-9926	VE-3305-2424-166	Dlngr	329-9925
						XL-90	AmrAir	329-9927

Cat No. 329-9875
- 2nd / Final Stage
- Side Engagement Holes
- 24 x 24 x 12 nominal
- 23.4 x 23.4 x 12 actual
- Galvanized Frame
- 1/4" Gaskets Ea. Face
- Microglass Media
- 2500 CFM Nom. Cap.
- 2µ Retention
- 0.9" W.C. initial.

Cat No. 329-9876
- 1st Stage
- 20 x 25 x 2 nominal
- 19.5 x 24.5 x 1.875 actual
- Galvanized Frame
- Optional Media's
  - #5 Polyester (add K5)
  - #200 Wire Mesh (add K200)
- 2000 CFM Nom. Cap.
- 10-750µ Retention
- 0.9" W.C. initial.

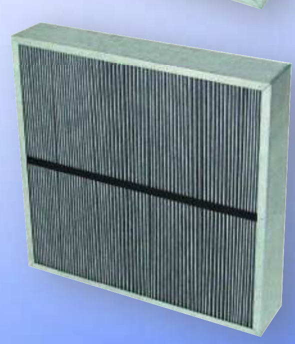
Cat No. 329-9925
- 3rd / Final Stage
- 24 x 24 x 12 nominal
- 23.4 x 23.4 x 12 actual
- Galvanized Frame
- 1/4" Gaskets Ea. Face
- Microglass Media
- 1250 CFM Nom. Cap.
- 0.3µ Retention
- 1" W.C. initial

Cat No. 329-9926
- 2nd Stage
- 24 x 24 x 6 nominal
- 23.4 x 23.4 x 6.25 actual
- Galvanized Frame
- 1/4" Gaskets Ea. Face
- Micrglass Media
- 1250 CFM Nom. Cap.
- 5µ Retention
- 0.5" W.C. initial

Cat No. 329-9927
- 2nd / Final Stage
- 24 x 24 x 12 nominal
- 23.4 x 23.4 x 12 actual
- Galvanized Frame
- 1/4" Gaskets Ea. Face
- Microglass Media
- 2500 CFM Nom. Cap.
- 2µ Retention
- 0.9" W.C. initial

Cat No. 329-9928
- 1st Stage
- 24 x 24 x 1.5 nominal
- 23.5 x 23.5 x 1.5 actual
- Galvanized Frame
- Polyester Media
- Washable / Reusable
- 2500 CFM Nom. Cap.
- 10-25µ Retention
- 0.6" W.C. initial

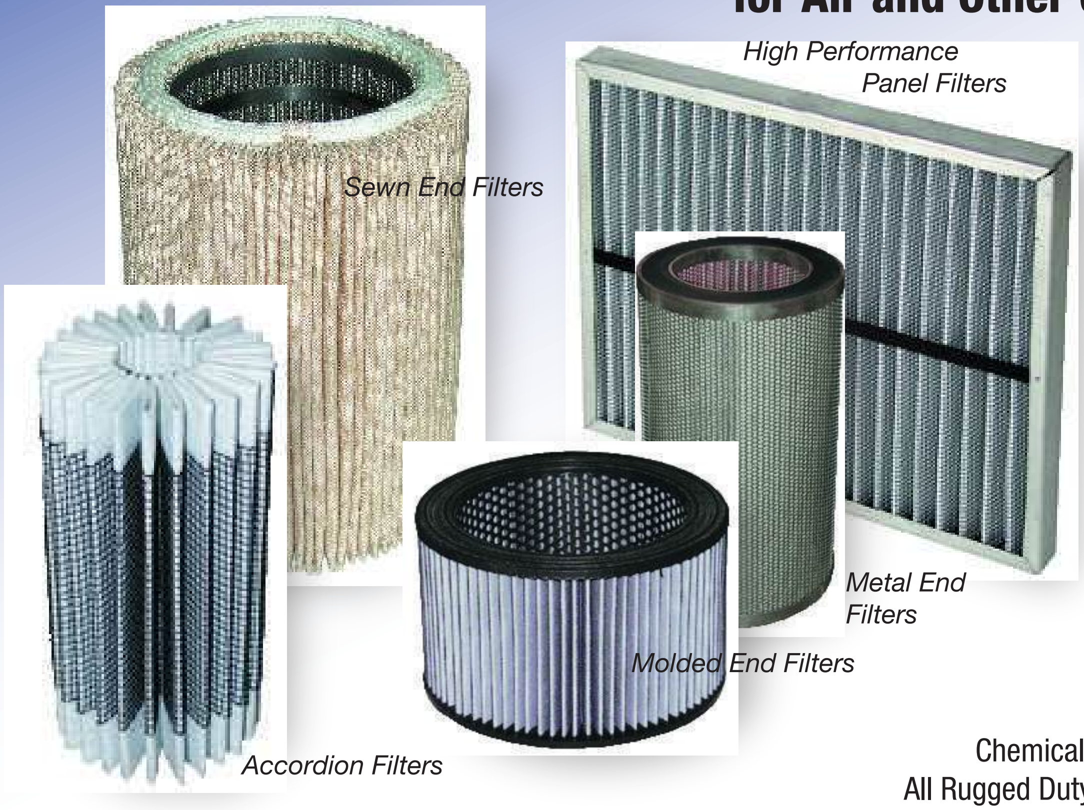
Cat No. 329-9929
- 1st or Single Stage
- 24 x 24 x 2 nominal
- 23.5 x 23.5 x 2 actual
- Galvanized Frame
- Polyester Media
- Washable / Reusable
- 2500 CFM Nom. Cap.
- 4-10µ Retention
- 1.1" W.C. initial



Cat No. 329-9784
- 1st or Single Stage
- 24 x 24 x 4 nominal
- 23.5 x 23.5 x 3.625 actual
- Galvanized Frame
- Various Medias
- 3000 CFM Nom. Cap.
- 10µ Retention
- 1" W.C. initial

SparksFilters Filter Elements

for Air and Other Gas Service:



High Performance Panel Filters

Sewn End Filters

Metal End Filters

Molded End Filters

Accordion Filters

Replacements for:

- Air Compressors
- Blowers
- Turbines
- Vacuum Pumps

- High Temp. Services
- Stock & Custom Designs to meet your Application.

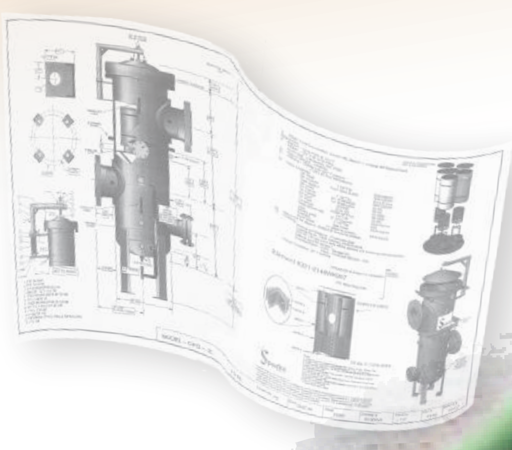
Plant Air
Food Processing
Power Generation
Cement Production
Pneumatic Conveying
Water Treatment
Waste Water Treatment
Chemical & Petro Chemical Processing
All Rugged Duty Air or Gas Filtration Services

Pressure & Vacuum Filters

for Air, Nat. Gas, or Other Gas Service:

- ASME Code
- All Connection Types & Sizes
- Design Pressures to 1480 psi (& higher)
- Rugged Steel, Stainless Steel, & other Alloy Construction

This ASME code certified coalescing filter removes mists and solid contamination from natural gas sent to an underground storage field in summer months, and later again when gas is retrieved from storage. The 20" unit employs a dual sump and requires no valving for seasonal flow change. It removed over 200 lbs of dirt and scale in its initial season of operation.



Installation Featured By ***World Business Review, On Location*** 2001/2002

Air/Gas Filters - Vacuum

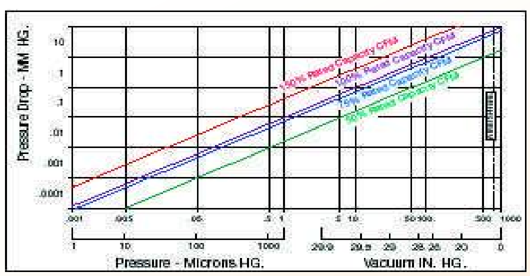
Series P20- Enameled Carbon Steel ◊ Series P22- 304 Stainless

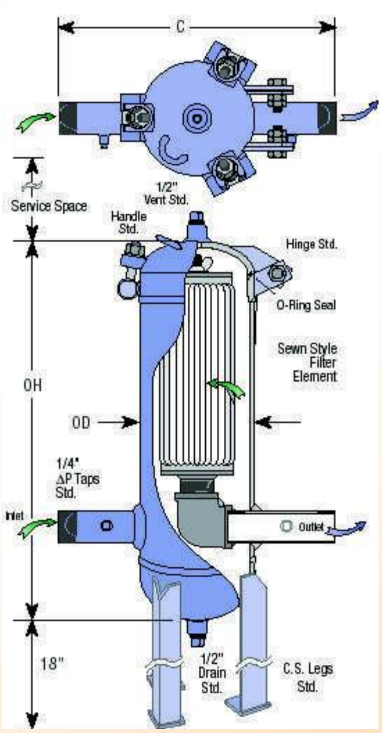
New solid urethane rubber end seals enable selecting either sewn end or rubber molded end style replacement filter elements.

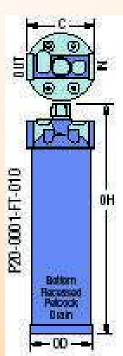
- **Low Pressure Drop**
- **Maximum Useful Filter Area and Dirt Holding Capacity**
- **Hinged Swing Bolt Closure, Easy Access, O Ring Seal**
- **304SS Throat Safety Cages, ΔP Taps Std.**
- **Rugged Enameled Steel Construction** Series P20 air and gas pipe line vacuum filters offer exceptional protection similar to our pressure vessels. They are fabricated from rugged enameled carbon steel. Any model can be modified at your request to more exactly fit your needs.
- **Standard Connection Sizes from 1" to 12"**
  NPT or raised face flanges (RF) in-line connections are std. Alternative orientations and sizes are available upon request. A hinged swing bolt closure is standard.
- **Recoverable Sewn or *Rubber* End Filter Elements.**
  Classic radial fin textile media elements are unsurpassed for low ΔP, dirt holding capacity and efficiency. They stop particulates before they can move downstream. Select either recoverable 10μ Sewn End Filter Elements or 10μ long life low cost polyurethane rubber molded end elements, as your needs dictate to remove 98% of all airborne dust and dirt. ***Unlike PVC or Plastisols, these polyurethane rubber ends have no phthalate elastomers to evaporate away in vacuum service.*** Our synthetic rubber is cross linked to further limit outgassing*. Dozens of alternative filter media are available for services having finer micron retentions, elevated temperatures and/or aggressive chemistry, see page 30. *If trace molecular contamination is of concern in your vaccum service, contact us for specifics.
- **Options:** Models P20-0087-RF-040 and larger include CS leg supports (add 18" to OH). Carbon steel support legs in any length, gauges and special finishes. Call for information on vessels having other pressure services, 304SS, or other materials of construction. See pgs. 14 & 15 for more options and a worksheet to fax back to us.

Housing Model No. (for 304SS, chg P20 to P22)	100% Pump Discpl. CFM	Conn. Size	Conn Type	Cover Style	Approx. Dimens, in.				Wgt. Lbs.	Select Filter Element	
					OD	OH	C	Serv. Space		Sewn End 10μ	*New* Rubber End 10μ
*P20-0001-FT-010	26	1"	FPT	T-Top	3½	14¼	4	12	7	320-0443K5	321-2789K5
P20-0084-MT-015	100	1.5"	MPT	Swg Blt	6⅝	28	16	16	80	320-0525K5	321-3235K5
P20-0085-MT-020	160	2"	MPT	Swg Blt	8⅝	30	16	16	100	320-0526K5	321-3236K5
P20-0086-MT-030	260	3"	MPT	Swg Blt	8⅝	39	20	24	175	320-0527K5	321-3237K5
P20-0087-RF-040	450	4"	Flg	Swg Blt	10¾	43	20	24	225	320-0528K5	321-3238K5
P20-0088-RF-060	850	6"	Flg	Swg Blt	12¾	52	24	26	300	320-0529K5	321-3239K5
P20-0089-RF-080	1800	8"	Flg	Swg Blt	16	67	28	36	500	320-0530K5	321-3240K5
P20-0090-RF-100	2800	10"	Flg	Swg Blt	20	74	32	36	850	320-0531K5	321-3241K5
P20-0091-RF-120	4000	12"	Flg	Swg Blt	24	82	36	39	1100	320-0532K5	321-3242K5

*ΔP Taps Not Available.







Welded steel wall and pre-fabricated closures yield for superior strength, closure seal, and corrosion allowance vs. lesser drawn shell designs offered by others.

Vacuum Conversion Table

Conversion of pressures between 760 and 0 torr

Vacuum Range	torr (mm Hg @ 32°F)	psia	Inches Hg @ 32°F	% Vacuum
Low	760	14.7	29.9	0
	750	14.5	29.5	1.3
	735	14.2	28.9	1.9
	700	13.5	27.8	7.9
	600	11.6	23.6	21
	500	9.7	19.7	34
	400	7.7	15.7	47
	380	7.3	15	50
	300	5.8	11.8	61
	200	3.9	7.85	74
	150	2.9	5.91	80
	100	1.93	3.94	87
Intermediate	90	1.74	3.54	88
	80	1.55	3.15	89.5
	70	1.35	2.76	90.8
	60	1.16	2.36	92.1
	51	1	2.03	93
	50	0.97	1.97	93.5
	40	0.77	1.57	94.8
	30	0.58	1.18	96.1
	25.4	0.49	1	96.6
	20	0.39	0.785	97.4
	10	0.193	0.394	98.7
	7.6	0.147	0.299	99
	1	0.019	0.039	99.9
High *	0.75	0.015	0.03	99.9
	0.1	0.0019	0.0039	99.99
	0.01	0.00019	0.0004	99.999
	0	0	0	100

* *Recommend **optional** 32 RMS machined flange surfaces.*

Air/Gas Filters - to 175 psig @ -20 to 200°F

Series P20- Enameled Carbon Steel ◊ Series P22- 304 Stainless

- **Intake Air Flows to 70,000 SCFM Std.**
- **ASME U Stamp Std., Nat'l. Board Registered**
- **Low Pressure Drop**
- **Maximum Useful Filter Area and Dirt Holding Capacity**
- **Hinged Swing Bolt Closure, Easy Access, O Ring Seal**
- **304SS Throat Safety Cages Standard**
- **ΔP Taps Standard**
- **Rugged Enameled Steel or 304SS Construction**

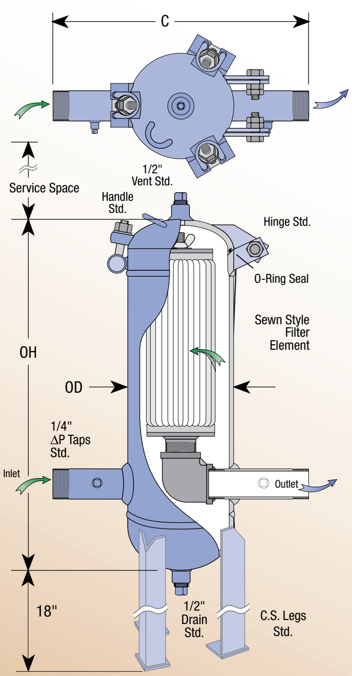
These air and gas pipe line filters offer exceptional protection for compressed gas pipelines, air dryers, pneumatic controls, and other pipeline equipment. They are fabricated from rugged enameled carbon steel (series P20) or 304SS (series P22), designed, constructed, and stamped, in accordance with ASME Boiler and Pressure Vessel Code requirements for unfired pressure vessels. Any model can be readily modified at your request to more exactly fit your needs.

- **Standard Connection Sizes from 1" to 12"**

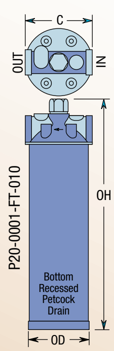
NPT or raised face flanges (RF) in-line connections are std. Alt. orientations & sizes are available upon request. A hinged swing bolt closure is standard on models P20-0002 & larger.

- **Recoverable Sewn or *Rubber* End Filter Elements.**

Classic radial fin textile media elements are unsurpassed for low ΔP, dirt holding capacity and efficiency. They stop particulates before they can move downstream. Select either recoverable 10µ Sewn End Filter Elements or 10µ long life low cost urethane rubber molded end elements, as your needs dictate to remove 98% of all airborne dust and dirt. Unlike PVC, our urethane rubber ends have no phthalate elastomers or other plasticizers. Dozens of alternative filter media are available for services having finer micron retentions, elevated temperatures and/or aggressive
chemistry, see page 30.



- **Options:** Models P20-0202-RF-030 and larger include CS leg supports. (add 18" to OH) Carbon steel support legs in any length, gauges, and special finishes, are optional on any model. Call for information on vessels having other pressure services, 304SS, or other materials of construction. See pgs. 14 & 15 for more options and a worksheet to fax back to us.

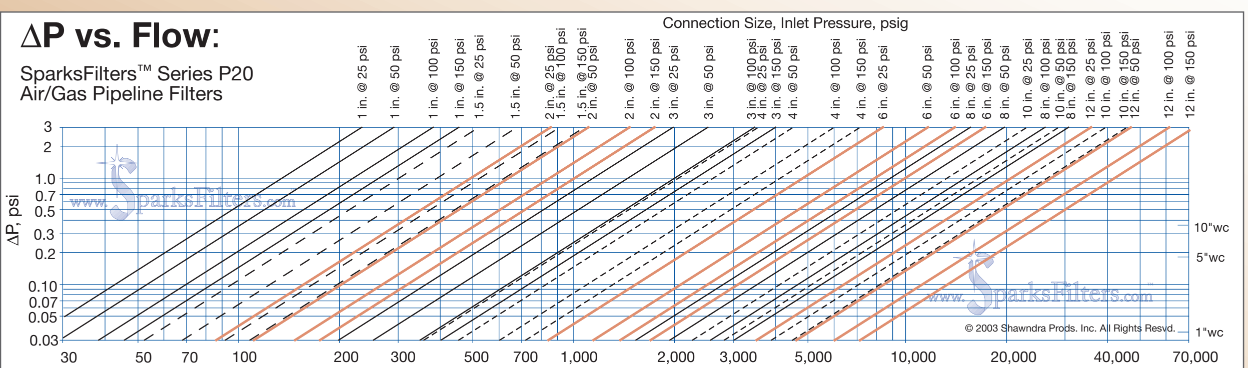


Housing Model No. (for 304SS, chg P20 to P22)	Conn. Size	Conn. Type	Cover Style	Dimensions in Inches				Wgt. Lbs.	Select Filter Element	
				OD	OH	C	Serv.		Sewn End	*New* Rubber
*P20-0001-FT-010	1"	FPT	T-Top	3½	14¼	4	12	7	320-0443K5	321-2789K5
P20-0200-MT-015	1.5"	MPT	Swg Blt	6⅝	28	16	16	80	320-0525K5	321-3235K5
P20-0201-MT-020	2"	MPT	Swg Blt	8⅝	30	16	16	110	320-0526K5	321-3236K5
P20-0202-RF-030	3"	Flg.	Swg Blt	8⅝	39	20	24	130	320-0527K5	321-3237K5
P20-0203-RF-040	4"	Flg.	Swg Blt	10¾	43	20	24	200	320-0528K5	321-3238K5
P20-0204-RF-060	6"	Flg..	Swg Blt	12¾	52	24	26	280	320-0529K5	321-3239K5
P20-0205-RF-080	8"	Flg.	Swg Blt	16	67	28	36	475	320-0530K5	321-3240K5
P20-0206-RF-100	10"	Flg.	Swg Blt	20	74	32	36	750	320-0531K5	321-3241K5
P20-0207-RF-120	12"	Flg.	Swg Blt	24	82	36	39	900	320-0532K5	321-3242K5

***ASME Code Stamp not available on this model, ΔP Taps Not Available MAWP=250 psig @ 200°F*

ΔP vs. Flow:

SparksFilters™ Series P20
Air/Gas Pipeline Filters



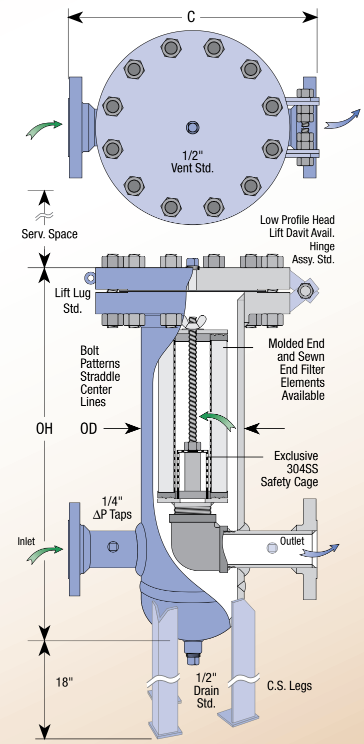
1. For service at temp. other than 60°F, multiply ΔP by: (460 + (oper. °F)) / 520
2. ΔP is chgd. proportionally with gas gravity, i.e. nat. gas ΔP 60% that of air.

Air/Gas Filters - to 1480 psig

1480psig @ -20 to 100°F
1350psig @ -20 to 200°F

SparksFilters™ **Series P50- Enameled Carbon Steel ◊ Series P52- 304 Stainless**

- **Intake Air Flows to 200,000 SCFM Std.**
- **ASME U Stamp Std., Nat'l. Board Registered**
- **Low Pressure Drop, Maximum Filter Area and Dirt Capacity**
- **Hinged Flange, Lift Lug, and 18" CS leg supports Std.**
- **Service Access W/O Breaking Connections**
- **304SS Throat Safety Cages and ΔP Taps Std.**
- **Rugged Enameled Steel or 304SS Construction** These air and gas pipe line filters offer exceptional protection for compressed gas pipelines, air dryers, pneumatic controls, and other pipeline equipment. They are fabricated from rugged enameled carbon steel (series P50) or 304SS (series P52), designed, constructed, and stamped, in accordance with ASME Boiler and Pressure Vessel Code requirements for unfired pressure vessels. Any model can be readily modified at your request to more exactly fit your needs.
- **Connection Sizes from 1" to 12"**
  Raised face flanges (RF) in-line connections are std. Alternative orientations and sizes are available upon request. A simple blind flange closure assembly is standard on all models.
- **Recoverable Sewn or *Rubber* End Filter Elements.**
  Classic radial fin textile media elements are unsurpassed for low ΔP, dirt holding capacity and efficiency. They stop particulates before they can move downstream. Select either recoverable 10μ Sewn End Filter Elements or 10μ long life low cost urethane rubber molded end elements, as your needs dictate to remove 98% of all airborne dust and dirt. Unlike PVC, our urethane rubber ends have no phthalate elastomers or other plastisizers. Dozens of alternative filter media are available for services having finer micron retentions, elevated temperatures and/or aggressive chemistry, see page 30.
- **Options:**
  Carbon steel support legs in any length, (18" CS leg supports are std. on all P50 and P52 models), gauges, special finishes, and head lift assemblies are optional on any model. Call for information on vessels having other pressure services, 304SS, or other materials of construction. See pgs. 14 & 15 for more options and a worksheet to fax back to us.

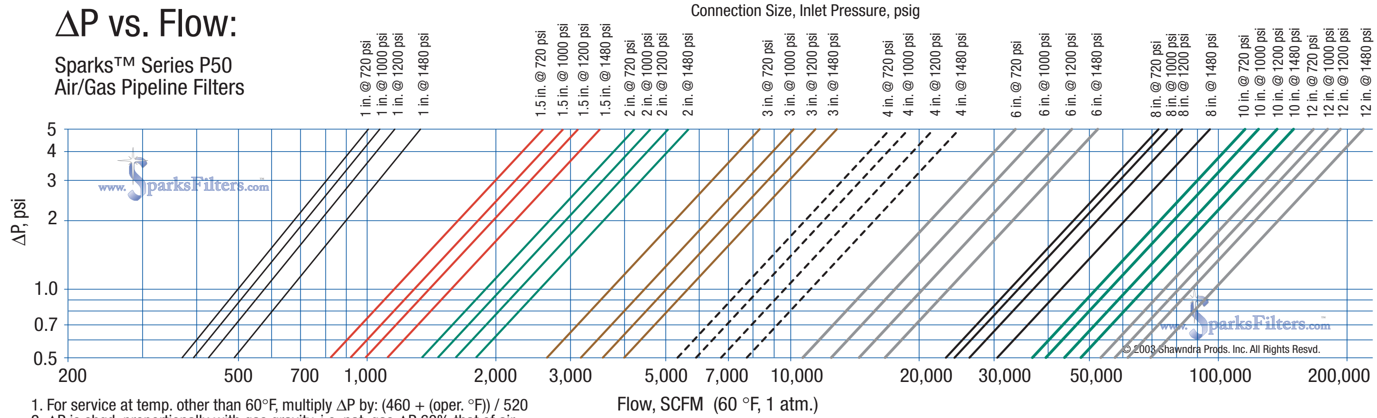


Housing Model No.	Conn. Size	Conn. Type	Cover Style	Dimensions in Inches				Wgt. Lbs.	Select Filter Elements	
				OD	OH	C	Serv. Space		Sewn End 10μ	Molded End 10μ
P50-0200-RF-010	1"	Flg.	Blind Flg.	6⅝	24	18	16	350	320-0365K5	321-1439K5
P50-0201-RF-015	1½"	Flg.	Blind Flg.	6⅝	28	18	16	400	320-0525K5	321-3235K5
P50-0202-RF-020	2"	Flg.	Blind Flg.	8⅝	28	18	16	450	320-0526K5	321-3236K5
P50-0203-RF-030	3"	Flg.	Blind Flg.	8⅝	39	24	24	500	320-0527K5	321-3237K5
P50-0204-RF-040	4"	Flg.	Blind Flg.	10¾	42	28	24	1000	320-0528K5	321-3238K5
P50-0205-RF-060	6"	Flg.	Blind Flg.	12¾	51	30	26	1200	320-0529K5	321-3239K5
P50-0206-RF-080	8"	Flg.	Blind Flg.	16	66	32	36	1800	320-0530K5	321-3240K5
P50-0208-RF-100	10"	Flg.	Blind Flg.	20	76	36	36	2600	320-0531K5	321-3241K5
P50-0209-RF-120	12"	Flg.	Blind Flg.	24	78	42	39	3600	320-0532K5	321-3242K5

Select filter elements with solid molded rubber ends or sewn ends with textile gaskets.

ΔP vs. Flow:

Sparks™ Series P50
Air/Gas Pipeline Filters



1. For service at temp. other than 60°F, multiply ΔP by: (460 + (oper. °F)) / 520
2. ΔP is chgd. proportionally with gas gravity, i.e. nat. gas ΔP 60% that of air.

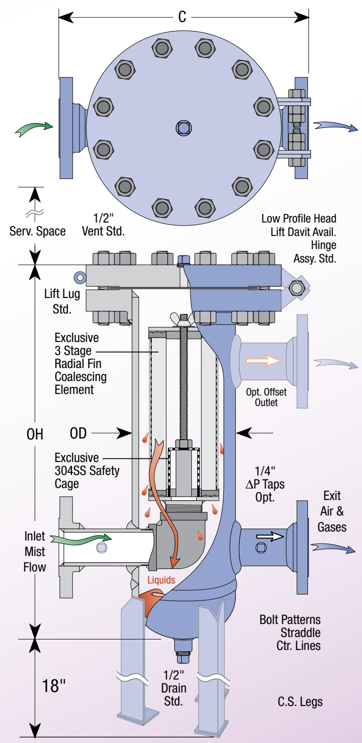
Coalescing Filters - to 1480 psig

1480psig @ -20 to 100°F
1350psig @ -20 to 200°F

Series R50- Enameled Carbon Steel ◊ Series R52- 304 Stainless

- **Intake Air Flows to 200,000 SCFM Std.**
- **ASME U Stamp Std., Nat'l. Board Registered**
- **Pleated Element Design, Exceptional Useful Filter Area**
- **Low ΔP, High Flow**
- **Hinged Flange w/Lift Lug Std.**
- **Service Access W/O Breaking Connections**
- **304SS Throat Safety Cages and ΔP Taps Std.**
- **Rugged Enameled Steel or 304SS Construction**
  Series R50 coalescing pipeline filters are fabricated from rugged enameled carbon steel, (R52 are 304SS), designed, constructed, and stamped in accordance with ASME Boiler and Pressure Vessel Code requirements for unfired pressure vessels. Any model can be modified to more exactly fit your needs.
- **Connection Sizes from 1" to 12"**
  Raised face flanges (RF) in-line connections are std. Alternative orientations and sizes are available upon request. A simple blind flange closure assembly is standard on all models.
- **Coalescing Filter Media.** Sparks™ #907 media is composed of microfine borosilicate glass fibers bonded with phenolic resin. Together with a textile prefilter and a final drain layer, these pleated elements are remarkably effective at coalescing fine entrained oil and aqueous vapor mist from air/gas flows with very low ΔP. Experience has demonstrated high removal (over 90%) in dealing with 1.0 to 0.3μ aerosols. ***Other optional filter media such as #926 exceeds 95% removals.*** Individual performance will vary with the specific viscosity and vapor pressure of liquid contaminates.
- **Options:**
  Carbon steel support legs in any length, (18" CS leg supports are std. on all R50 and R52 models), gauges, special finishes, and head lift assemblies are optional on any model. Call for information on vessels having other pressure services, 304SS, or other materials of construction.

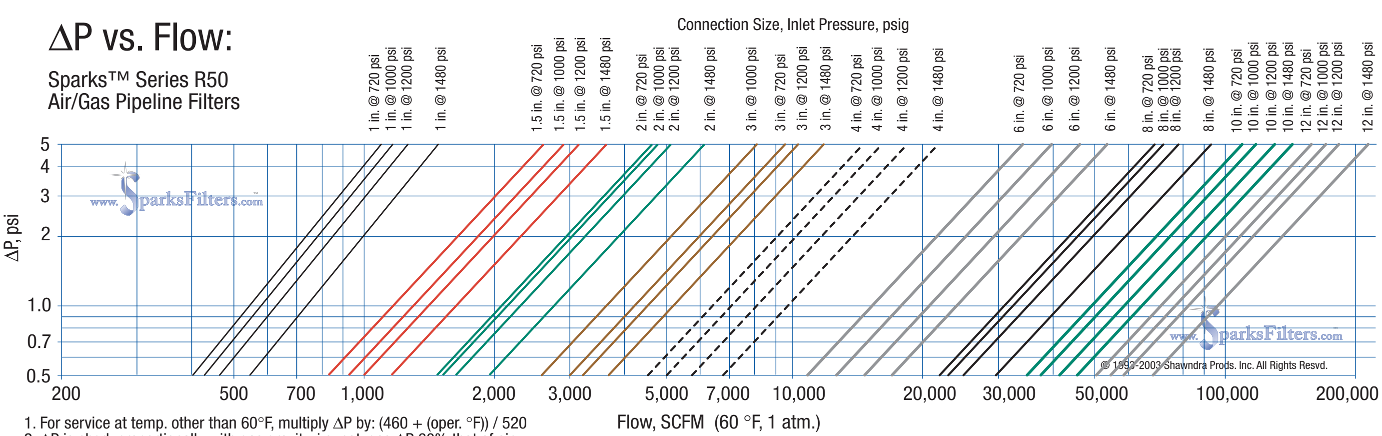
See pgs. 14 & 15 for more options and a worksheet to fax back to us.



Housing Model No.	Conn. Size	Conn. Type	Cover Style	Dimensions in Inches				Wgt. Lbs.	Order Element No.
				OD	OH	C	Serv. Space		
R50-0200-RF-010	1"	Flg.	Blind Flg.	6⅝	24	18	16	350	321-1439WK907
R50-0201-RF-015	1½"	Flg.	Blind Flg.	6⅝	28	18	16	400	321-3235WK907
R50-0202-RF-020	2"	Flg.	Blind Flg.	8⅝	28	18	16	450	321-3236WK907
R50-0203-RF-030	3"	Flg.	Blind Flg.	8⅝	39	24	24	500	321-3237WK907
R50-0204-RF-040	4"	Flg.	Blind Flg.	10¾	42	28	24	1000	321-3238WK907
R50-0205-RF-060	6"	Flg.	Blind Flg.	12¾	51	30	26	1200	321-3239WK907
R50-0206-RF-080	8"	Flg.	Blind Flg.	16	66	32	36	1800	321-3240WK907
R50-0207-RF-100	10"	Flg.	Blind Flg.	20	76	36	36	2600	321-3241WK907
R50-0208-RF-120	12"	Flg.	Blind Flg.	24	78	42	39	3600	321-3242WK907

ΔP vs. Flow:

Sparks™ Series R50 Air/Gas Pipeline Filters



1. For service at temp. other than 60°F, multiply ΔP by: (460 + (oper. °F)) / 520
2. ΔP is chgd. proportionally with gas gravity, i.e. nat. gas ΔP 60% that of air.

Side Exhaust Mist Eliminating Filters

Series H20 - Enameled Steel • Series H22 - 304 Stainless Steel

SparksFilters™

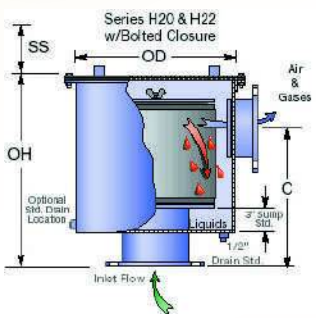
Access Handles Standard on Models with OD 12" and greater.

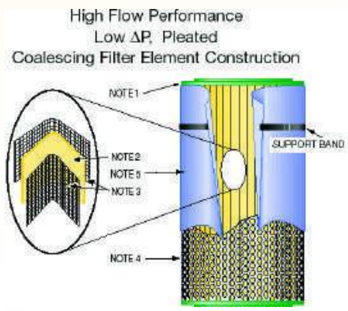
- **Connection Sizes 1 ¼" to 24"**
- **All Steel Construction.** • **Options:** ΔP Taps, ΔP gauges, angle legs

  Series H20 (enamel steel) and Series H22 (304SS) each employ a bolt seal closure with bottom inlet and side exhaust permitting remote vapor routing if desired.
- **Rugged Design - 15 psid** @ -20 to 200°F (Non Code)
- **304 Stainless Steel Throat Safety Cages Standard**
- **Three Styles of Exhaust Elements Available**

  Classic Two Stage Microglass Exhaust Elements are time-honored performers. However, their non-pleated cylindrical construction can generate a higher pressure drop (typ. ΔP 2 psi clean) and requires a larger filter housing to accept the physically large exhaust element.

  Newer High Flow Low ΔP Pleated Microglass Elements have permited a less expensive more compact housing design together with a proven much lower pressure drop. Their extended filter area effectively retains both particulates and mist droplets with removal performance to 0.3μ The extended filter area provides superior dirt and particulate retention capacities. Oil & Gas Resistant Ends with double track seals out perform metal end style elements.





1. Oil & Gas Resistant Ends
2. MicroGlass/Synthetic composite media.
3. Upstream/Downstream epoxy coated screen jackets.
4. Perforated steel core; Perforated outer shell (corrosion control coated).
5. Outer foam drain wrap.

High Flow, Low ΔP, Pleated Microglass Exhaust Elements with field removable foam drain wrap.

Classic Dual Stage Microglass Exhaust Filter Element.

Enameled Steel Housing Cat. No.	304SS Housing Cat. No.	Typ. Flow CFM	Std. Connection Size	Type	Approximate Dimensions, Inches OH	OD	C	SS	Wgt. lbs.	Select One Filter Element Below: Classic Filter Element	0.3 μ High Flow Pleated Element	≤ 0.3 μ High Performance Pleated Elem.
H20-0001-MT-012	H22-0001-MT-012	40	1.25"	M	13.5	6.63	9	5	23		321-2118WK907	321-2118WK927
H20-0012-MT-012	H22-0012-MT-012	40	1.25"	M	9.25	6.63	6	6	12	321-1890		
H20-0001-MT-015	H22-0001-MT-015	50	1.5"	M	13.5	6.63	9	5	23		321-2118WK907	321-2118WK927
H20-0013-MT-015	H22-0013-MT-015	50	1.5"	M	10.5	6.63	7	7	13	321-1864		
H20-0014-MT-015	H22-0014-MT-015	50	1.5"	M	13.5	10.75	9	10	33	321-1895		
H20-0002-MT-020	H22-0002-MT-020	100	2"	M	16	6.63	12	8	24		321-2119WK907	321-2119WK927
H20-0015-MT-020	H22-0015-MT-020	100	2"	M	17.5	16	12	12	75	321-1893		
H20-0003-MT-030	H22-0003-MT-030	250	3"	M	24	8.63	20	16	38		321-2120WK907	321-2120WK927
H20-0016-MT-030	H22-0016-MT-030	250	3"	M	23.5	20	17	17	125	321-1884		
H20-0004-MT-040	H22-0004-MT-040	350	4"	M	24	10.75	20	16	55		321-2121WK907	321-2121WK927
H20-0017-MT-040	H22-0017-MT-040	350	4"	M	23.5	24	17	17	175	321-1894		
H20-0005-FF-050	H22-0005-FF-050	700	5"	Flg	40	12.75	34	25	85		321-2122WK907	321-2122WK927
H20-0018-FF-050	H22-0018-FF-050	700	5"	Flg	30.5	24	24	24	195	321-1891		
H20-0005-FF-060	H22-0005-FF-060	1000	6"	Flg	40	12.75	34	25	90		321-2122WK907	321-2122WK927
H20-0019-FF-060	H22-0019-FF-060	1000	6"	Flg	30.5	24	24	24	200	321-1891		
H20-0006-FF-080	H22-0006-FF-080	1500	8"	Flg	40	16	34	25	120		321-2123WK907	321-2123WK927
H20-0007-FF-100	H22-0007-FF-100	2400	10"	Flg	44	20	34	25	160		321-2124WK907	321-2124WK927
H20-0008-FF-120	H22-0008-FF-120	3400	12"	Flg	44	24	34	25	200		321-2125WK907	321-2125WK927
H20-0009-FF-160	H22-0009-FF-160	5400	16"	Flg	48	32	34	21	350		321-2126WK907	321-2126WK927
H20-0010-FF-200	H22-0010-FF-200	8500	20"	Flg	48	36	34	25	450		321-2127WK907	321-2127WK927
H20-0011-FF-240	H22-0011-FF-240	12,000	24"	Flg	48	44	34	25	650		321-2128WK907	321-2128WK927

Industrial Process Filters

We Modify and Custom Design Filter Components For Solutions That Fit Your Specific Needs!

Series P32 w/all flanged connections including ports for level gauge, relief valve, drain valve.

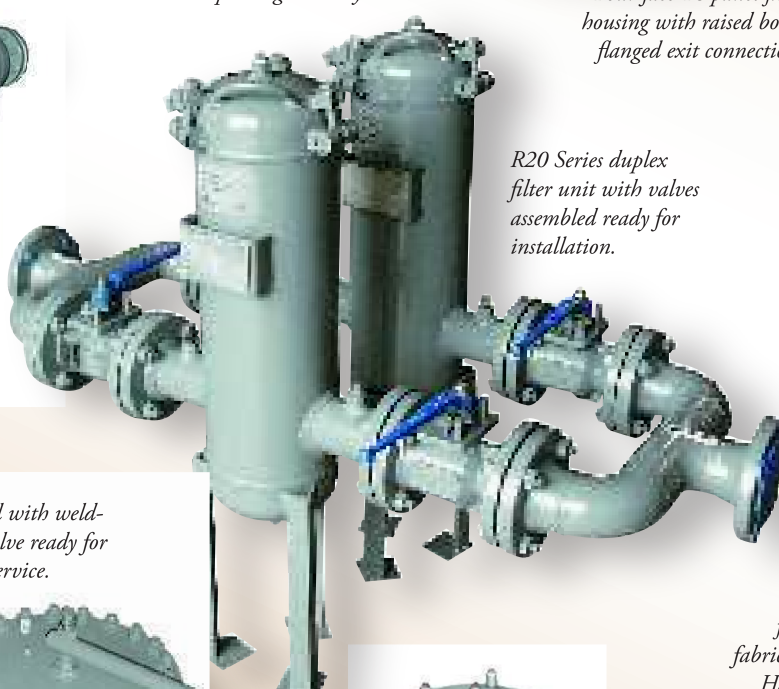
Inverted series R50. Leg design allows unencumbered service.

Optional single ΔP gauge installed across inlet/outlet improving accuracy.

Four face 16 panel filter housing with raised bottom flanged exit connection.

Single face S10-0001 panel filter w/o weather hood.

Horiz. (ASME Code) sump tank for R Series vessels.



R20 Series duplex filter unit with valves assembled ready for installation.

P-Series filter vessel fabricated from Hastelloy® C and 304SS.

P-Series vessel with welded transfer valve ready for liquid filter service.

Stainless steel chemical process holding vessel.

Stainless steel bladder style accumulator.